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Dear Editor,

I am pleased to submit my manuscript, “*Coupled Thermal Activation and Autocatalytic Cure Kinetics: A Theoretical and Computational Study of Delayed Reactive Transformation*,” for consideration as a theoretical/computational research article in Macromolecular Theory and Simulations. The manuscript has been posted to ChemRxiv as a preprint: <https://doi.org/10.26434/chemrxiv.15008366/v1>.

The manuscript formulates and computationally analyzes a fully coupled zero-dimensional dynamical model of a thermally activated, autocatalytically curing system: a lumped thermal energy balance (including reaction exotherm), a physically motivated Arrhenius release-relaxation process, and Kamal–Sourour autocatalytic cure kinetics. Four dimensionless groups — Λ (release-to-reaction timescale ratio), Θ (thermal-to-reaction timescale ratio), ψ (reduced exothermicity), and Ar (activation-sensitivity number) — organize the parameter space. Every reported comparison is objective rather than assumed: a first-order control's rate constant is root-solved to match the coupled system's t_{50} exactly, and a full 60-point Pareto sweep shows the coupled system is non-dominated by first-order controls on premature conversion and completion time, and the extended analysis described below tests this claim against stronger controls and a storage-stability objective.

This version adds four analyses, each designed to test a conclusion rather than to enlarge the model. A capsule-population model derives the gate width from a distribution of melting temperatures ($w_{\text{eff}} \approx 0.55\sigma$), so that the leakage prediction acquires a measurable input, the width of the DSC melting endotherm. A storage-stability objective, which the earlier draft did not measure, is added and the constant-rate first-order control is replaced by Arrhenius controls that share the energy balance: none of 547 admissible designs of four other mechanisms dominates the coupled system, whereas some detuned designs of the coupled family itself dominate its illustrative reference point, which is stated explicitly. A design rule gives the largest admissible melting-point spread (about 6 °C for 1 % storage conversion at a 6 °C margin), and Sobol indices confirm the dominance of the kinetic activation energy. A table lists every claim with its evidential status, the assumption on which it rests and what would reverse it, and two further falsifiable predictions are added. All numbers are model outputs under illustrative parameters; no experimental data are presented.

This is a substantially revised version of a model I previously submitted elsewhere as an isothermal, prescribed-release treatment; that submission's review feedback — that thermal transients and exotherm were neglected, that the release function lacked physical justification, that the control comparison used an unjustified scaling factor, that the induction-time result was a structural consequence of the model rather than an independent prediction, and that a single dimensionless ratio could not organize the system — is addressed point-by-point by specific structural changes in this version: a genuine coupled energy balance, an Arrhenius-derived release process, objective control matching with a Pareto sweep, an explicit structural-versus-predicted distinction, and a two-parameter regime map replacing the single ratio. Numerical integration is verified by comparing fixed-step Euler and RK4 against an adaptive RK45 reference (sub-0.1% agreement), and joint parameter uncertainty across nine kinetic, release, and thermal parameters is propagated via a vectorized Monte Carlo scheme ($N=20,000$).

The manuscript is explicit throughout about what is modeled versus established: no experimental or material-specific data is presented, and five falsifiable predictions — centered on a finite-width thermal gate's measurable trade-off between premature conversion and completion time — are each paired with a concrete dynamic-DSC or rheo-DSC validation protocol for future experimental testing.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. I am the sole author and have approved the manuscript for submission. I have no conflicts of interest to declare, and this work received no external funding.

The complete simulation code (integration, sweeps, Monte Carlo, and figure generation) is openly available at <https://github.com/sandlerleon/autolatch-thermal-gating> (MIT license) and archived with a permanent DOI on Zenodo: <https://doi.org/10.5281/zenodo.22073392>. The manuscript itself is additionally archived as a preprint on Zenodo: <https://doi.org/10.5281/zenodo.22073390>. The present version is archived as <https://doi.org/10.5281/zenodo.23202124> (code) and <https://doi.org/10.5281/zenodo.23202125> (manuscript).

Thank you for considering this manuscript. I would be glad to suggest potential reviewers or provide any further information the editorial office may need.

Sincerely,

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